

Combinatorics – 8

Permutations, binary recursions and log-concavity

Homework

24 сентября 2026 года

Theoretical minimum. For $\sigma \in S_n$ one has $\sum_{\sigma} q^{\text{inv } \sigma} = \prod_{k \leq n} (1 + \dots + q^{k-1})$ by the Lehmer code and $\sum_{\sigma} x^{\text{cyc } \sigma} = x(x+1) \cdots (x+n-1)$. The Eulerian numbers $A(n, k)$ count permutations with k descents; inserting n gives $A(n, k) = (k+1)A(n-1, k) + (n-k)A(n-1, k-1)$, and $\sum_{m \geq 0} (m+1)^n t^m = A_n(t)/(1-t)^{n+1}$; excedances have the same distribution (Foata). Sign and cycle type are Abstract algebra – 3. A complete mapping of an abelian group G is a bijection σ with $x \mapsto x + \sigma(x)$ bijective; it forces $\sum_{g \in G} g = 0$, so $\mathbb{Z}_n$ has one iff n is odd. Functions given by $f(2n)$, $f(2n+1)$ are handled by induction along the binary expansion, e.g. $s_2(2n+1) = s_2(n) + 1$, $v_2(n!) = n - s_2(n)$, Stern's $s(2n) = s(n)$, $s(2n+1) = s(n) + s(n+1)$. A positive sequence with $a_k^2 \geq a_{k-1}a_{k+1}$ (log-concave) is unimodal, convolution preserves log-concavity, and real-rooted polynomials with non-negative coefficients have log-concave coefficients (Newton; Polynomials – 4). Parity: an involution of a finite set X has $|\text{Fix}| \equiv |X| \pmod{2}$.

1. Prove that a log-concave sequence of positive numbers is unimodal and that the convolution of two such sequences is log-concave. Let $B_n(t) = \sum_{m \geq 0} (m+1)^n t^m$; check that $B_{n+1}(t) = (tB_n(t))'$ and prove by Rolle's theorem that $A_n(t)$ has $n-1$ distinct negative roots. Deduce that the Eulerian numbers and the Stirling numbers $c(n, k)$ are log-concave in k , and locate the maximum of $A(n, k)$.
2. Let X be an arbitrary set and let f be a one-to-one function mapping X onto itself. Prove that there exist mappings $g_1, g_2: X \rightarrow X$ such that $f = g_1 \circ g_2$ and $g_1 \circ g_1 = \text{id} = g_2 \circ g_2$, where id denotes the identity mapping on X . (IMC 1997)
3. Let N_n denote the number of ordered n -tuples of positive integers

$(a_1, a_2, \dots, a_n)$ such that $1/a_1 + 1/a_2 + \dots + 1/a_n = 1$. Determine whether N_{10} is even or odd. (Putnam 1997 A5)

4. Define a sequence by $a_0 = 1$, together with the rules $a_{2n+1} = a_n$ and $a_{2n+2} = a_n + a_{n+1}$ for each integer $n \geq 0$. Prove that every positive rational number appears in the set

$$\left\{ \frac{a_{n-1}}{a_n} : n \geq 1 \right\} = \left\{ \frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{1}{3}, \frac{3}{2}, \dots \right\}.$$

(Here $a_n = s(n+1)$ is Stern's sequence.) (Putnam 2002 A5)

5. For every positive integer n denote by D_n the number of permutations $(x_1, \dots, x_n)$ of $(1, 2, \dots, n)$ such that $x_j \neq j$ for every $1 \leq j \leq n$. For $1 \leq k \leq n/2$ denote by $\Delta(n, k)$ the number of permutations $(x_1, \dots, x_n)$ of $(1, 2, \dots, n)$ such that $x_i = k + i$ for every $1 \leq i \leq k$ and $x_j \neq j$ for every $1 \leq j \leq n$. Prove that

$$\Delta(n, k) = \sum_{i=0}^{k-1} \binom{k-1}{i} \frac{D_{(n+1)-(k+i)}}{n - (k+i)}.$$

(IMC 2014)

6. A sequence $a_1, a_2, \dots, a_n$ is called bright if all its members are positive integers and the numbers $a_1 + 1, a_2 + 2, \dots, a_n + n$ give all possible remainders upon division by n . Let the sequence $a_1, a_2, \dots, a_{2025}$ be bright with $a_1 = 1, a_2 = 2, \dots, a_9 = 9$. What is the minimal possible value of $a_1 + a_2 + \dots + a_{2025}$? (IMC 2025, shortlist)

7. Consider a circular necklace with 2013 beads. Each bead can be painted either white or green. A painting of the necklace is called good if among any 21 successive beads there is at least one green bead. Prove that the number of good paintings of the necklace is odd. (Two paintings that differ on some beads, but can be obtained from each other by rotating or flipping the necklace, are counted as different paintings.) (IMC 2013)

8. A positive integer $n > 1$ is called good if the set $\{1, 2, 3, \dots, 3n\}$ can be divided into ordered triples (a_i, b_i, c_i) , $1 \leq i \leq n$, such that all n quadratic equations $a_i x^2 + b_i x + c_i = 0$ have a common root. Prove

that there exist infinitely many positive integers n that are good and also infinitely many positive integers that are not good. (IMC 2025, shortlist)

9. We say that a positive real number d is good if there exists an infinite sequence $a_1, a_2, a_3, \dots \in (0, d)$ such that for each n , the points $a_1, \dots, a_n$ partition the interval $[0, d]$ into segments of length at most $1/n$ each. Find $\sup\{d \mid d \text{ is good}\}$. (IMC 2021)
10. Let $n \geq 2$ be an integer. For a sequence $s = (s_1, \dots, s_{n-1})$ where each $s_i = \pm 1$, let $f(s)$ be the number of permutations $(a_1, \dots, a_n)$ of $\{1, 2, \dots, n\}$ such that $s_i(a_{i+1} - a_i) > 0$ for all i . For each n , determine the sequences s for which $f(s)$ is maximal. (Putnam 2025 A5)